Assignment 11

1.

a. Class 1 : {1}
Class 2 : {7, 8}
Class 3 : {3, 4, 5, 6}
Class 4 : {2}

b. ~~Transient → {1, 2, 5, 6}~~
~~Recurrent → {4, 7, 8, 3}~~

c. Periods of all states

State 1 : 1 − 1 = 1 → Aperiodic
State 2 : 0 → Neither Periodic nor Aperiodic → Bridging / Transient state
State 3 : 3 − 4 − 3 = 2
3 − 4 − 5 − 6 − 3 = 4
~~gcd (2, 4) = 2~~
3 − 4 − 5 − 6 − 6 − 3 = 5
d(i) = gcd (2, 4, 5) = 1 → Aperiodic

State 4 : 4 − 3 − 4 = 2
4 − 5 − 6 − 3 − 4 = 4
4 − 5 − 6 − 6 − 3 − 4 = 5
d(i) = gcd (2, 4, 5) = 1 → Aperiodic

State 5 : 5 − 6 − 3 − 4 − 5 = 4
5 − 6 − 6 − 3 − 4 − 5 = 5
5 − 6 − 3 − 4 − 3 − 4 − 5 = 6
5 − 6 − 6 − 3 − 4 − 3 − 4 − 5 = 7
d(i) = gcd (4, 5, 6, 7) = 1 → Aperiodic

State 6 : 6 − 6 = 1
6 − 3 − 4 − 5 − 6 = 4
6 − 6 − 3 − 4 − 5 − 6 = 5
6 − 3 − 4 − 3 − 4 − 5 − 6 = 6
6 − 6 − 3 − 4 − 3 − 4 − 5 − 6 = 7
d(i) = gcd (1, 4, 5, 6, 7) = 1 → Aperiodic

State 7 : 7 - 8 - 7 = 2

7 - 8 - 8 - 7 = 3

$d(i) = \gcd(2,3) = 1 \rightarrow$ Aperiodic

State 8 : 8 - 8 = 1

8 - 7 - 8 = 2

$d(i) = \gcd(1,2) = 1 \rightarrow$ Aperiodic

b. Transient state - Start from i & probablity of returning back is greater than 0.

→ 1, 2, 5, 6

Recurrent state - Start from i & definately return back to i

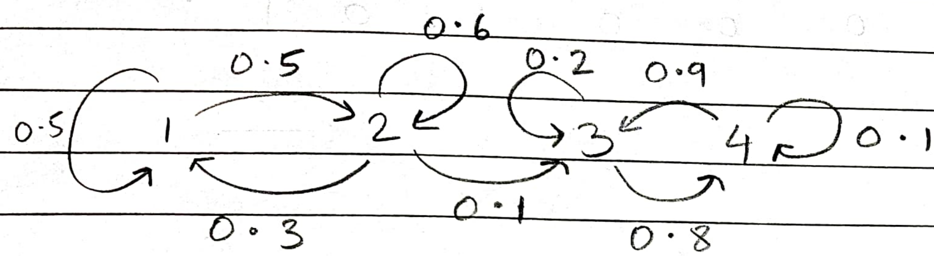
- 3, 4, 7, 8

d. TPM :

$$
\begin{array}{c|cccccccc}
 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\
\hline
1 & 0.5 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 \\
2 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0.5 & 0 \\
3 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
4 & 0 & 0 & 0.5 & 0 & 0.5 & 0 & 0 & 0 \\
5 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
6 & 0 & 0 & 0.5 & 0 & 0 & 0.5 & 0 & 0 \\
7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
8 & 0 & 0 & 0 & 0 & 0 & 0 & 0.75 & 0.25
\end{array}
$$

2.(i)

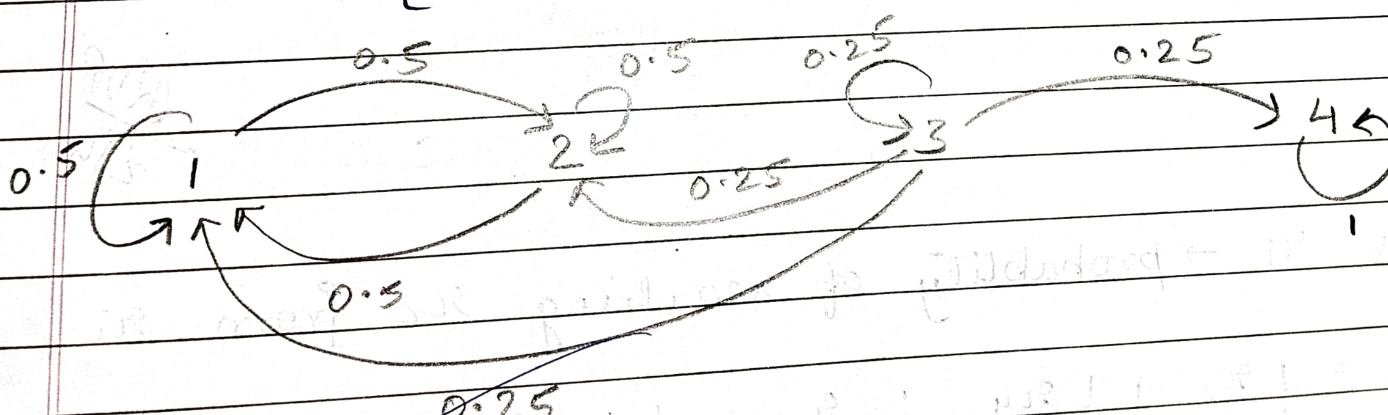
$$P_1 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0.5 & 0.5 & 0 & 0 \\ 2 & 0.3 & 0.6 & 0.1 & 0 \\ 3 & 0 & 0 & 0.2 & 0.8 \\ 4 & 0 & 0 & 0.9 & 0.1 \end{array}$$



Class 1 : {1, 2}
Class 2 : {3, 4}

(ii) $P_2 =$

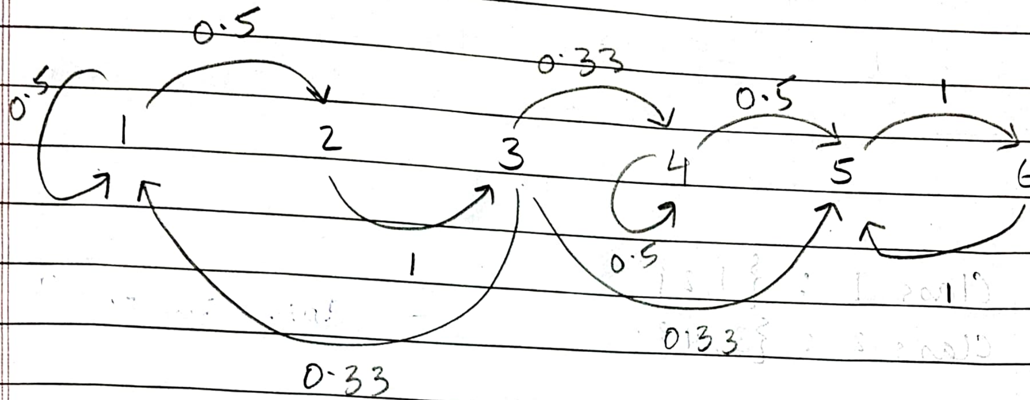
$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0.5 & 0.5 & 0 & 0 \\ 2 & 0.5 & 0.5 & 0 & 0 \\ 3 & 0.25 & 0.25 & 0.25 & 0.25 \\ 4 & 0 & 0 & 0 & 1 \end{array}$$



Class 1 : {1, 2}
Class 2 : {3}
Class 3 : {4}

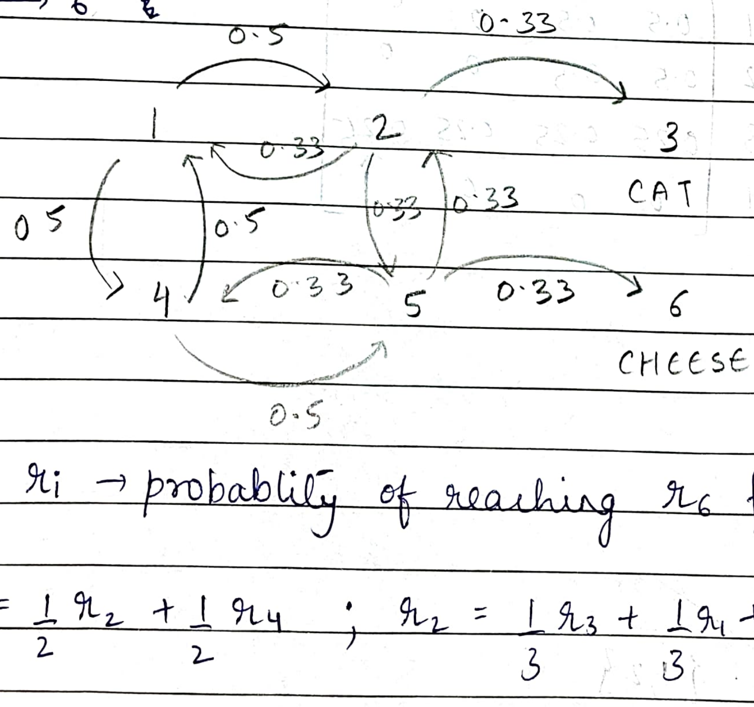
3.

$$P = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0.33 & 0 & 0 & 0.33 & 0.33 & 0 \\ 0 & 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$



$5 \longleftrightarrow 6$

4.



RVP
2/4

Let $r_i \rightarrow$ probablity of reaching r_6 from r_i

$r_1 = \frac{1}{2} r_2 + \frac{1}{2} r_4$; $r_2 = \frac{1}{3} r_3 + \frac{1}{3} r_1 + \frac{1}{3} r_5$;

$r_3 = 0$; $r_4 = \frac{1}{2} r_1 + \frac{1}{2} r_5$; $r_5 = \frac{1}{3} r_2 + \frac{1}{3} r_4 + \frac{1}{3} r_6$

$\boxed{r_5 = 63.6\%}$

$r_6 = 1 \longrightarrow$ Substituting values of r_3, r_6 :

$r_1 = \frac{1}{2} r_2 + \frac{1}{2} r_4$; $r_2 = \frac{1}{3} r_1 + \frac{1}{3} r_5$; $r_4 = \frac{1}{2} r_1 + \frac{1}{2} r_5$;

$r_5 = \frac{1}{3} r_2 + \frac{1}{3} r_4 + \frac{1}{3} \longrightarrow \boxed{r_1 = 45.5\%, \ r_2 = 36.4\%, \ r_4 = 54.5\%}$